

A natural account of loss aversion

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0.1 Basic idea

The current model also provides a natural explanation of loss aversion in prospect theory. In a nutshell, the absolute subjective value (i.e., the utility) of a change in endowment is greater for losses than for gains - according to some meta-analyses about 1.995 times as large [Brown et al., 2024] (but see below). While this observation has been considered “the most significant contribution of psychology to behavioral economics” ([Kahneman, 2011], p. 300), it is usually treated as a quirk of human decision making by fitting arbitrary parameters to human performance [Tversky and Kahneman, 1992, Abdellaoui et al., 2007, Booij and van de Kuilen, 2009].

Instead, we propose that loss aversion is an automatic consequence of our biophysically-rooted decision model. Specifically, we propose that, with respect to an endowment E , the subjective desirability (i.e., the utility) of a positive change C tracks the subjective certainty that the change is a gain, which, in turn tracks the Weber ratio $\frac{E+C}{E}$. Conversely, the subjective *undesirability* of a negative change (i.e., the utility) tracks the subjective certainty that the change is a loss, which, in turn, tracks the Weber ratio $\frac{E}{E+C}$ (for negative C 's).

The Weber ratio for losses $\frac{E}{E-|C|}$ is strictly greater than that for gains $\frac{E+|C|}{E}$, where $|C|$ is the absolute value of C .¹ Since the acceptability function is monotonic, the absolute utility of losses is thus strictly greater than that of gains. In other words, loss aversion is a basic consequence of magnitude processing, and thus of any system with a fixed precision and multiplicative gain control [Endress and Szabó, 2020].

This idea is illustrated in Figure 1. Figure 1a plots the subjective value as a function of relative changes with respect to the endowment between -99% and 99%, and shows the typical S-shaped curve. Figure 1b plots the relative utility for loss vs. gains for a given change, i.e. $\frac{|Utility_{Loss}|}{|Utility_{Gain}|}$. The relative utility is a measure of loss aversion. For example, if $\frac{|Utility_{Loss}|}{|Utility_{Gain}|} = 2$ for a given change, the change is twice as aversive as a loss as it would be positive as a gain. Figure 1b shows that the relative utility is always greater than 1, and thus shows loss aversion for all uncertainty parameters w . Further, we make the novel prediction that the relative utility of losses (compared to gains) should increase as a function of change (see below), a prediction that receives support from different studies [Abdellaoui et al., 2007, Booij and van de Kuilen, 2009].

0.2 Comparison to [Tversky and Kahneman, 1992]

In Figure 2, we show that our model can mimic the functional form assumed by [Tversky and Kahneman, 1992]. We use their parameterization (utility = $|C|^\alpha - \lambda|C|^\beta$, with $\lambda = 2.25$ and $\alpha = \beta = 0.88$, but see below). We thus set the w parameter to a large value to make it quasi-linear as in their function ($w = 10$). Given that both models use arbitrary units, we multiplied the output of the Tversky and Kahneman model so that the maximum y value coincide for both models. As shown in Figure 2, both models behave fairly similarly.

0.3 Variability of λ estimates

While [Tversky and Kahneman, 1992] proposed a λ value of 2.25, there is considerable variability across studies, with some metanalyses finding a value of 1.995 [Brown et al., 2024], others of 1.31 [Walasek et al.,

¹Proof: Assume that $\frac{E}{E-|C|} < \frac{E+|C|}{E}$. It follows that $E^2 < (E+|C|)(E-|C|)$, and thus that $|C|^2 < 0$, which is a contradiction.

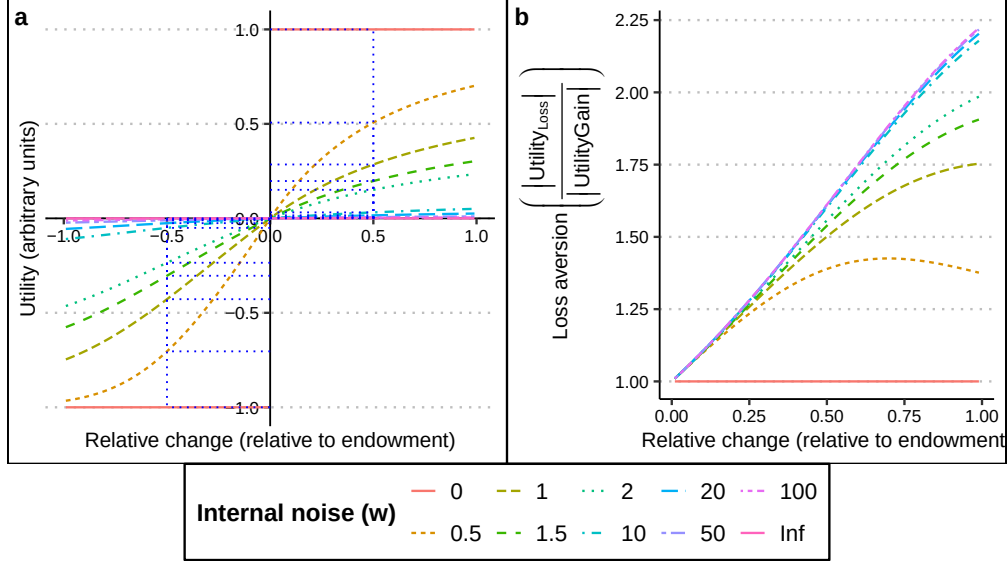


Figure 1: Illustration of loss aversion as a consequence of magnitude processing. (a) Subjective utility as a function of relative gains or losses with respect to an endowment, for different values of the uncertainty parameter w . The S-shaped curve emerges naturally from the asymmetry of the Weber ratios for gains and losses. Dotted blue lines mark the utility values at a change of $\pm 50\%$, illustrating that the absolute utility of a 50% loss exceeds that of a 50% gain. (b) Ratio of absolute utilities for losses vs. gains, $\frac{|Utility_{Loss}|}{|Utility_{Gain}|}$, as a function of relative change. Values greater than 1 indicate loss aversion. Note that loss aversion exceeds 1 for all values of w and increases with the size of the change.

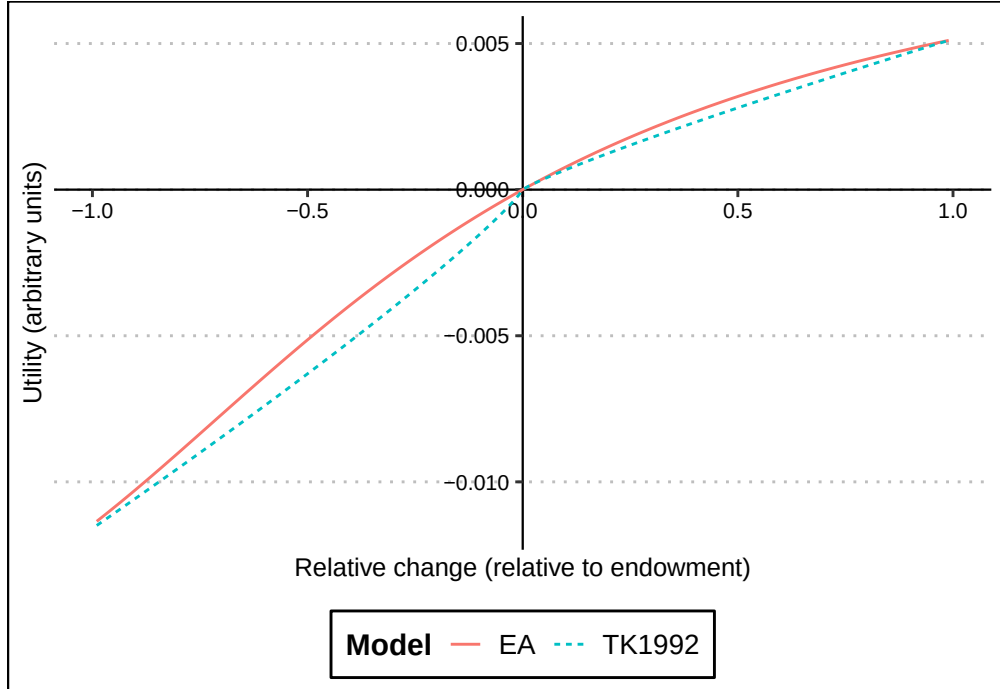


Figure 2: Comparison of the current model (EA) and the Tversky and Kahneman (1992) model, with parameterization $\lambda = 2.25$, $\alpha = \beta = 0.88$. The w parameter was set to a large value ($w = 10$) to approximate the quasi-linear behavior of the Tversky and Kahneman (1992) function. Because both models use arbitrary units, the output of the Tversky and Kahneman (1992) model was scaled so that the maxima of both functions coincide. The two models produce closely overlapping value functions across the full range of relative changes.

2024] and yet others of 1.07 [Yechiam and Zeif, 2025]. Further, methodological differences seems to affect the magnitude of loss aversion, including how the parameter is estimated, the population [Brown et al., 2024], and the size of the magnitude of the gain/loss [Abdellaoui et al., 2007, Booij and van de Kuilen, 2009], with some authors even proposing that loss aversion mainly arises in experiments where gains are larger than losses and where gains and losses are presented in an ordered fashion [Yechiam and Zeif, 2025].

While our simple conceptual model is not intended to capture such variability, variability is expected for a variety of reasons. For example, and as mentioned above, we predict that loss aversion depends on the relative change. Likewise, given that number processing shows substantial individual differences in the w parameter [Halberda et al., 2008], we would expect similar individual differences in loss aversion. To the extent that observers retain numeric information over trials, we would also expect to see ordering effects like those in [Yechiam and Zeif, 2025], for example through anchoring effects across trials. However, further research is needed to map our model to such differences in loss aversion.

0.4 Integration of probabilistic information

In standard prospect theory, utilities are integrated with probability information. However, given that humans use approximate representations of numbers even when these numbers are presented as Arabic numerals [Moyer and Landauer, 1967, Buckley and Gillman, 1974, Hinrichs et al., 1981, Dehaene et al., 1990], the assumption that probabilistic information is represented at face value is almost certainly false (see also [Barretto-García et al., 2023, Khaw et al., 2020] for evidence that risk taking is related to number processing precision). As a result, to construct a biophysically grounded derivation of prospect theory, more work is need to understand how probabilistic information is represented, and how the (presumably) noisy representations of probability information is integrated with the noisy representations of value.

0.5 Monotonicity behavior [We will probably remove this section eventually, but I keep it in case I have a better idea for the proof]

In this section, we investigate whether the relative utility of losses vs. gains is growing. With the acceptability given by $1 + \log_2 \Phi \left(\frac{R-1}{\sqrt{2}w\sqrt{(R^2+1)}} \right)$, we get

$$G(C) = \frac{1 + \log_2 \Phi \left(\frac{\frac{E}{E-C} - 1}{\sqrt{2}w\sqrt{\left(\frac{E}{E-C}\right)^2 + 1}} \right)}{1 + \log_2 \Phi \left(\frac{\frac{E+C}{E} - 1}{\sqrt{2}w\sqrt{\left(\frac{E+C}{E}\right)^2 + 1}} \right)} = \frac{1 + \log_2 \Phi \left(\frac{C}{\sqrt{2}w\sqrt{E^2 + (E-C)^2}} \right)}{1 + \log_2 \Phi \left(\frac{C}{\sqrt{2}w\sqrt{(E+C)^2 + E^2}} \right)}$$

where we assume that C is non-negative. While the derivative is cumbersome to calculate, is easy to see that $G(C) > 1$ for all C 's between 0 and E , and that $G(C)$ must be locally growing for at least some C in the neighborhood of 0.

In fact, for $C = 0$, L'Hôpital's Rule shows that $G(C)$ converges to 1. For all $0 < C < E$, the Weber ratio in the numerator is strictly greater than that in the denominator. As both $\log_2$ and Φ are strictly monotonic, $G(C)$ is greater than 1 for all $0 < C < E$.

As a result, the Mean Value Theorem implies that, for each $0 < C < E$, there exists an C^* for which $G'(C^*)$ is strictly positive. While this does not imply that G grows monotonically, it does imply that the one can find C^* 's where G' is positive in arbitrarily small neighborhoods of 0, and that the average growth around $C = 0$ is positive.

To avoid differentiating an unwieldy function, we consider the difference in utilities.

$$H(C) = 1 + \log_2 \Phi \left(\frac{\frac{E}{E-C} - 1}{\sqrt{2}w\sqrt{\left(\frac{E}{E-C}\right)^2 + 1}} \right) - \left(1 + \log_2 \Phi \left(\frac{\frac{E+C}{E} - 1}{\sqrt{2}w\sqrt{\left(\frac{E+C}{E}\right)^2 + 1}} \right) \right) = \log_2 \Phi \left(\frac{C}{\sqrt{2}w\sqrt{E^2 + (E-C)^2}} \right) - \log_2 \Phi \left(\frac{C}{\sqrt{2}w\sqrt{(E+C)^2 + E^2}} \right)$$

References

- Mohammed Abdellaoui, Han Bleichrodt, and Corina Paraschiv. Loss aversion under prospect theory: A parameter-free measurement. *Management Science*, 53(10):1659–1674, 2007. doi: 10.1287/mnsc.1070.0711. URL <https://doi.org/10.1287/mnsc.1070.0711>.
- Miguel Barretto-García, Gilles de Hollander, Marcus Grueschow, Rafael Polanía, Michael Woodford, and Christian C. Ruff. Individual risk attitudes arise from noise in neurocognitive magnitude representations. *Nature human behaviour*, 7:1551–1567, September 2023. ISSN 2397-3374. doi: 10.1038/s41562-023-01643-4.
- Adam S. Booij and Gijs van de Kuilen. A parameter-free analysis of the utility of money for the general population under prospect theory. *Journal of Economic Psychology*, 30(4):651–666, 2009. ISSN 0167-4870. doi: <https://doi.org/10.1016/j.joep.2009.05.004>. URL <https://www.sciencedirect.com/science/article/pii/S0167487009000555>.
- Alexander L. Brown, Taisuke Imai, Ferdinand M. Vieider, and Colin F. Camerer. Meta-analysis of empirical estimates of loss aversion. *Journal of Economic Literature*, 62(2):485–516, June 2024. doi: 10.1257/jel.20221698. URL <https://www.aeaweb.org/articles?id=10.1257/jel.20221698>.
- P. B. Buckley and C. B. Gillman. Comparisons of digits and dot patterns. *Journal of experimental psychology*, 103:1131–1136, December 1974. doi: 10.1037/h0037361.
- S. Dehaene, E. Dupoux, and J. Mehler. Is numerical comparison digital? Analogical and symbolic effects in two-digit number comparison. *J Exp Psychol Hum Percept Perform*, 16(3):626–41, 1990.
- Ansgar D. Endress and S. Szabó. Sequential presentation protects memory from catastrophic interference. *Cognit Sci*, 44(5), 2020. doi: 10.1111/cogs.12828.
- Justin Halberda, Michèle M M Mazzocco, and Lisa Feigenson. Individual differences in non-verbal number acuity correlate with maths achievement. *Nature*, 455:665–668, 2008. ISSN 1476-4687. doi: 10.1038/nature07246.
- James V. Hinrichs, Dale S. Yurko, and Jing mei Hu. Two-digit number comparison: Use of place information. *Journal of Experimental Psychology: Human Perception and Performance*, 7(4):890–901, aug 1981. doi: 10.1037/0096-1523.7.4.890.
- Daniel Kahneman. *Thinking, fast and slow*. Penguin books, London, UK, 2011. ISBN 978-0-141-03357-0.
- Mel Win Khaw, Ziang Li, and Michael Woodford. Cognitive Imprecision and Small-Stakes Risk Aversion. *The Review of Economic Studies*, 88(4):1979–2013, 08 2020. doi: 10.1093/restud/rdaa044.
- R. S. Moyer and T. K. Landauer. Time required for judgements of numerical inequality. *Nature*, 215:1519–1520, September 1967. doi: 10.1038/2151519a0.
- Amos Tversky and Daniel Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5:297–323, 1992.
- Lukasz Walasek, Timothy L. Mullett, and Neil Stewart. A meta-analysis of loss aversion in risky contexts. *Journal of Economic Psychology*, 103:102740, 2024. doi: <https://doi.org/10.1016/j.joep.2024.102740>.
- Eldad Yechiam and Dana Zeif. Loss aversion is not robust: A re-meta-analysis. *Journal of Economic Psychology*, 107:102801, 2025. ISSN 0167-4870. doi: <https://doi.org/10.1016/j.joep.2025.102801>. URL <https://www.sciencedirect.com/science/article/pii/S0167487025000133>.